Introduction

CSC 1105 Mathematics for Computer Science

Denish Azamuke, PhD

Office: CoCIS, Block B, Level 3

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Welcome to the Course

- This course introduces mathematical ideas that support core areas of computer science.
- We shall focus on reasoning, abstraction, calculation, and interpretation.
- The course connects mathematics to computing areas such as machine learning, graphics, interface design, robotics, image processing, and security.

Course Information

Item	Details
Course Name	Mathematics for Computer Science
Course Code	CSC 1105
Course Credit	4
Contact Hours	60
Year of Study	1
Semester	1
University	Department of Computer Science, Makerere University
Office	Block B, Level 3

Lecture Timetable

Class	Time	Venue
Day (CS-1)	Monday, 10:00–12:00	LLT 2C
	Tuesday, 10:00–12:00	LLT 5B
Evening (CS-EVE-1)	Monday, 15:00–17:00	LLT 5A
	Tuesday, 15:00–17:00	LLT 2C

Office consultation: CoCIS, Block B, Level 3.

Why Mathematics for Computer Science?

- Mathematics provides a language for describing computational problems precisely.
- It supports algorithm design, data analysis, modelling, simulation, graphics, and artificial intelligence.
- It helps computer scientists justify solutions instead of only implementing them.
- The goal is not memorisation, but mathematical thinking for computing.

Course Description

- The course builds a foundation and deep understanding of mathematical skills needed for computer science.
- It extends high-school mathematics with emphasis on concepts useful in computer science.
- Main application areas include image processing, machine learning, graphics, interface design, robotics, and security.

Course Objectives

By the end of the course, students should develop skills in:

1. Finding limits of functions.
2. Establishing continuity of functions.
3. Evaluating finite and infinite sequences and progressions.
4. Performing operations in matrix algebra.
5. Analysing and interpreting vector spaces.
6. Performing plane arithmetic.

Learning Outcomes

By the end of the course, students should be able to:

- Compute limits of polynomial, logarithmic, exponential, trigonometric, and mixed functions.
- Establish continuity and differentiability of functions.
- Calculate inverses, determinants, and transposes of matrices.
- Perform 2D and 3D transformations using matrices.
- Perform vector and vector-space algebra.
- Generate mathematical characteristics for vectors and planes.

Course Modules

Module	Topic	CH
Module 1	Limits and Continuity	10
Module 2	Progressions and Sequences	10
Module 3	Matrix Algebra	10
Module 4	Vectors and Vector Spaces	15
Module 5	Vectors and Planes	15

Module Highlights

- **Limits and Continuity:** limits, limit laws, L'Hopital's rule, differentiability, derivatives, and rates of change.
- **Progressions and Sequences:** arithmetic and geometric progressions, limits, bounds, and recurrence sequences.
- **Matrix Algebra:** rules, transposes, inverses, determinants, eigenvectors, kernels, transformations, and equations.
- **Vectors and Planes:** vector spaces, linear combinations, spanning sets, independence, orthogonality, lines, planes, angles, and distances.

Mode of Delivery

- The course will use learner-centred delivery through face-to-face and/or online classes.
- The instructor will introduce each topic and guide students on further reading.
- Student-centred activities such as discussions and forums will support learning.
- Students are expected to practise regularly, ask questions, and work through examples independently.

Assessment

- Each module will be assessed by at least one assignment and one test.
- Progressive assessment contributes **40%** of the final score.
- The final examination contributes **60%** of the final score.
- Consistent practice is essential because later topics build on earlier ones.

Study Materials

- Textbooks and lecture notes.
- Prepared tutorials and practice questions.
- Group discussions and online webinars.
- Recommended references:
  - Robert A. Beezer, *A First Course in Linear Algebra*.
  - Crowell and Slesnick, *Calculus with Analytic Geometry*.

How to Succeed in This Course

- Attend lectures and tutorials consistently.
- Attempt problems before looking at solutions.
- Form small study groups for discussion and peer explanation.
- Visit the office during consultation hours when you need support.
- Connect every mathematical idea to a computing problem.

Mathematics is not only about obtaining answers.

It is about learning how to reason clearly about problems.

Welcome to **CSC 1105 Mathematics for Computer Science**.